

need for human intervention, provides a valuable service and lowers the total cost of ownership, thus making PCs more valuable as information resources in our lives. You can see what autonomic computing has already created," asserts J. Gerry Purdy, Principal Analyst, MobileTrax.

One may also recall Fox Broadcasting's coverage of Super Bowl XXXVI in February 2002, wherein Fox's information technology team relied on self-diagnosing, self-healing IBM technology to supply viewers with statistics, fast facts and graphics. The on-site network at the Superdome helped Fox provide up-to-the-minute research, script and graphics preparation, and network management. The servers featured technologies from IBM's Project eLiza autonomic computing initiative to make computers self-diagnosing and self-healing. And the personal computers include a patented technology that helps computer users recover more easily from a crash due to a corrupted application or operating system, relying on data stored on a separate area of the hard drive.

Then, the ThinkPad of the future is expected to automatically establish the settings needed to connect to a network, enu-

How Grid Computing comes into play

IBM has been a leader in Grid computing. It is then but logical, to inspect the relationship of Grid computing with autonomic computing. Grid computing harnesses unused processing cycles of all computers in a network for solving problems too intensive for any stand-alone machine. The SETI (Search for Extraterrestrial Intelligence) @Home project is a familiar name in this regard. As federated systems emerge, new standards and policies will be required, and each element—processor, storage, network, software—becomes an autonomic computing element.

Meta Group views parallel processing as part of a larger body of processing concepts that we call 'federated computing'. This also includes peer-to-peer processing (as done by Gnutella and others), instant messaging and other systems that enable computers to work together without a

server in the middle. Federated computing helps break complex problems into discrete tasks, which can be farmed out to multiple machines.

If autonomic capabilities are to be extended across the Internet, you need a set of open protocols. These open protocols allow the various systems around the Internet that are participating in this autonomic initiative to be able to collaborate and share information, identify denial of service attacks, identify any failures, and start routing things around.

This is where Grid computing comes in, because it is the open protocols of the Grid running on every single system that enable 'intelligent' management capabilities to cut across different systems. Using these Grid protocols, IBM avers, it can realise this vision of creating, over time, an increasingly self-managing infrastructure.

merate the connection options and renew connections as users move around, without any user input.

IBM's investment in autonomic

info Sun Microsystems plans to introduce the N1 initiative, an OS for data centres that turns the whole network into a computer

computing includes Client Recovery and Rescue, which would help people recover data and continue some operations even after a catastrophic PC failure such as a broken hard drive. Another is Distributed Wireless Security Auditor (DWSA), which allows PCs in the same location to work together to detect dangerous 'rogue' security risks in wireless networks, saving hours of manpower. And a third, Instant Connections, automatically detects wireless and wired networks and configures the PC to work with them, saving people from having to

remember server addresses and wireless protocols. IBM expects to introduce some of these innovative technologies in 2003.

What some critics question is whether this is really the beginning of autonomic computing or mere business hype over a

few great features. The fact remains that complete autonomic systems do not yet exist.

Simply 'intelligent machines'?

Yes and no. In a sense there's nothing new about many of the elements of autonomic computing. Those working with complex adaptive systems, using ants, agents or evolution, or pursuing other biological metaphors such as artificial immune systems, embryonics and cybernetics, will find a familiar ring. One can also relate it to terms such as 'intelligent machine' and 'artificial intelligence'.

IBM, on its part, clarifies that if 'intelligent machine' means one that embodies human cognitive powers, the answer is no. But if that term is taken to mean systems that can adapt, learn and take over certain functions previously performed by humans, then autonomic computing does involve the idea of embedding this kind of intelligence in computing systems.

As for Artificial Intelligence, some



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